

Anterior abdominal wall

The anterior abdominal wall constitutes a hexagonal area defined superiorly by the costal margins and xiphoid process, laterally by the mid-axillary line, and inferiorly by the iliac crests, pubis and pubic symphysis. It is continuous with the posterior abdominal wall and paravertebral tissues, forming a flexible sheet of skin, muscle and connective tissue across the anterior and lateral aspects of the abdomen. These tissues also form the umbilicus and the inguinal canal, which connects the abdominal cavity to the scrotum in men or to the labia majora in women. The anterior abdominal wall maintains the shape of the abdomen and aids numerous physiological functions. However, its dysfunction is equally notable because hernia repair is the single most common operation performed by general surgeons.

SKIN AND SOFT TISSUE

The integument of the anterior abdominal wall comprises skin, soft tissues, and lymphatic and vascular structures, as well as segmental nerves. The outer layer is formed from the skin and subcutaneous fat. The skin is non-specialized and variably hirsute, depending on sex and race. All postpubertal individuals have some extension of the pubic hair on to the anterior abdominal wall skin, although this is commonly most pronounced in males, in whom the hair may extend almost up to the umbilicus in a triangular pattern. The subcutaneous fat of the abdominal wall is highly variable in thickness, depending in part on gender and adiposity.

SOFT TISSUE

Superficial fascia

The 'superficial fascia' of the abdominal wall lies between the dermis and the muscles, and is conventionally divided into a superficial fatty layer (Camper's fascia) and a deep membranous layer (Scarpa's fascia). In reality, there are three layers, with a further layer of adipose tissue deep to the membranous layer ([Lancerotto et al 2011](#)). These three layers are particularly well defined in the child. Lying within the superficial fascia are blood vessels, lymphatics, nerves and, in the region of the groin, superficial inguinal lymph nodes.

Superficial adipose layer

The superficial layer contains a variable amount of fat that is partitioned by fibrous septa connecting the dermis with the deeper membranous layer. Inferiorly, it is continuous with the superficial fascia of the thigh, and medially, it is continuous over the linea alba. In the male, this layer continues over the external genitalia, where it becomes thin and pale red, and contains very little adipose tissue. In the scrotum it also contains the smooth muscle fibres of the dartos muscle. In the female, it continues from the suprapubic region of the abdomen into the labia majora and perineum.

Membranous layer

The membranous layer is a variably developed entity composed of connective tissue and elastic fibres. In the adult, its thickness varies over the anterior abdominal wall, becoming thinner in the upper abdomen ([Lancerotto et al 2011](#)). Measured histologically, it is between 0.5 and 1 mm thick but it appears thicker on computed tomography (CT) scans ([Lancerotto et al 2011](#), [Chopra et al 2011](#)). It is loosely connected to the underlying external oblique aponeurosis and rectus sheath by oblique fibrous septa. Superiorly, it is continuous with the superficial fascia over the remainder of the trunk. In the midline, it is

adherent to the linea alba and pubic symphysis. Inferiorly, it fuses with the iliac crest, extends superficial to the inguinal ligament and fuses with the fascia lata at the inguinal flexure or skin crease of the thigh. In the male, it extends on to the dorsum of the penis, forming part of the superficial ligament of the penis, and on to the scrotum, where it becomes continuous with the membranous layer of superficial fascia of the perineum (Colles' fascia). In the female, it continues into the labia majora and is continuous with the fascia of the perineum.

In boys, the testis can frequently be retracted out of the scrotum into the loose areolar tissue between the membranous layer of superficial fascia over the inguinal canal and the external oblique aponeurosis. This 'space' is sometimes called the superficial inguinal pouch.

Deep adipose layer

The thickness of the deep adipose layer is more variable than the superficial fatty layer. It is thin or absent where the membranous layer fuses with bony prominences and the linea alba, and becomes markedly thick in the morbidly obese. Its adipocytes show different metabolic activities to those in the superficial adipose layer ([Chopra et al 2011](#)). Liposuction preferentially removes this layer of fat with relative preservation of the superficial adipose layer in order to avoid skin dimpling and other skin contour irregularities ([Markman and Barton 1987](#)).

Transversalis fascia

The transversalis fascia is a thin layer of connective tissue lying between the deep surface of transversus abdominis and the extraperitoneal fat. It is part of the general layer of thin fascia between the peritoneum and the abdominal wall. Posteriorly, it fuses with the anterior layer of the thoracolumbar fascia ([p. 1083](#)), and anteriorly, it forms a continuous sheet. Superiorly, it blends with the fascia covering the inferior surface of the diaphragm. Inferiorly, it is continuous with the iliac and pelvic parietal fasciae, and is attached to the iliac crest between the origins of transversus abdominis and iliacus, and to the posterior margin of the inguinal ligament between the anterior superior iliac spine and the femoral sheath. Medial to the femoral sheath it is thin and fused to the pubis behind the conjoint tendon. An inferior extension of the transversalis fascia forms the anterior part of the femoral sheath. The fascia displays a discrete thickening known as the iliopubic tract (also called the deep crural arch), which runs parallel to the inguinal ligament ([Teoh et al 1999](#)); it consists of transverse fibres that fan out laterally towards the anterior superior iliac spine to blend with the iliopsoas fascia and run medially behind the conjoint tendon to the pubic bone. The iliopubic tract is recognized as an important structure during open and laparoscopic inguinal hernia repair. A further thickening of the transversalis fascia, the interfoveolar ligament, runs inferior to the inguinal ligament at the medial margin of the deep inguinal ring; it may contain muscle fibres.

The transversalis fascia is prolonged as the internal spermatic fascia over the structures that pass through the deep inguinal ring (the testicular vessels and vas (ductus) deferens in the male and the round ligament of the uterus in the female).

Extraperitoneal connective tissue

The extraperitoneal connective tissue lying between the peritoneum and the fasciae lining the abdominal and pelvic cavities contains a variable amount of fat. The fat is especially abundant on the posterior wall of the abdomen around the kidneys (particularly in obese men) and scanty above the iliac crest and in much of the pelvis.

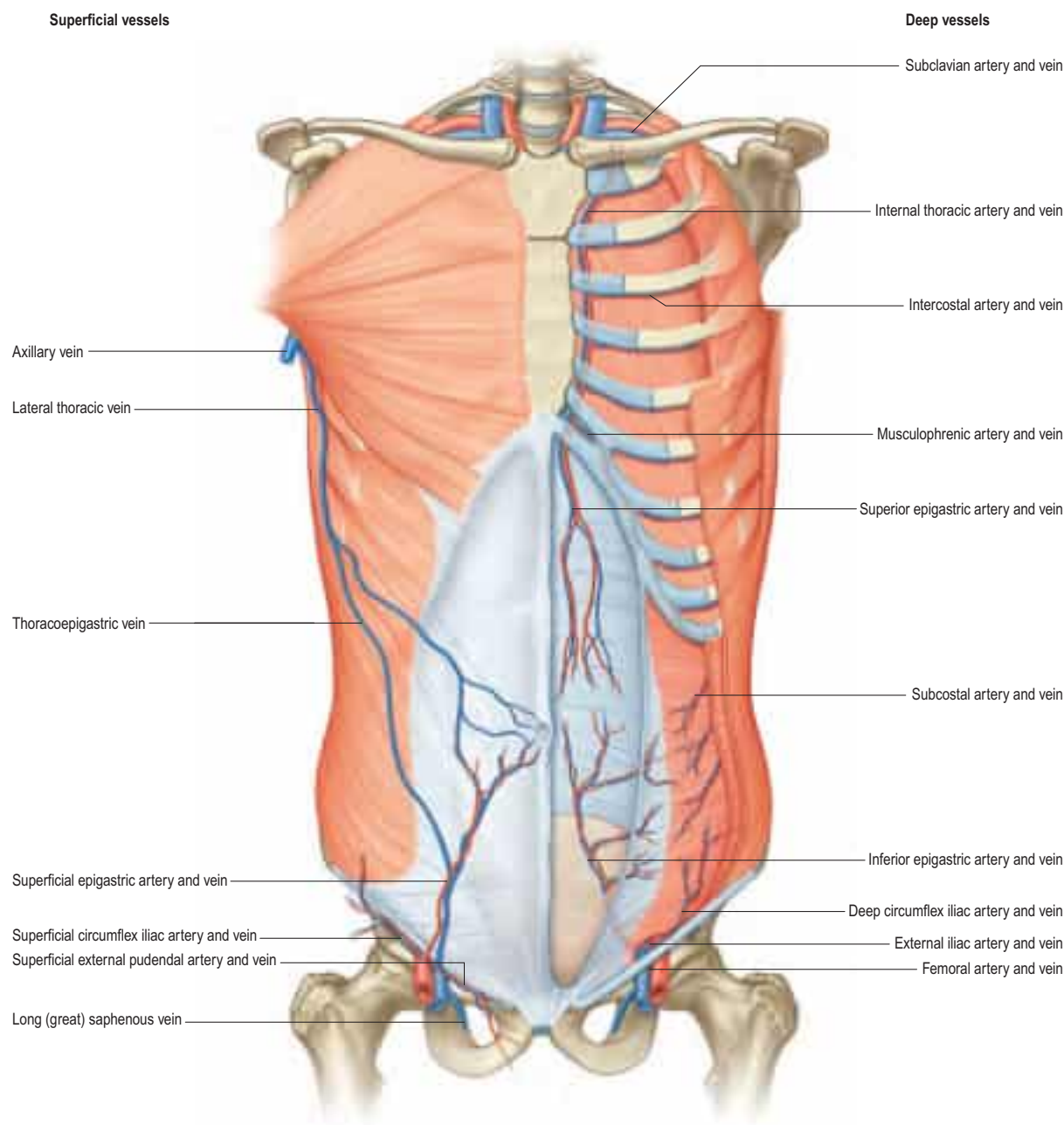


Fig. 61.1 The blood supply of the anterior abdominal wall.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

Understanding the blood supply of the abdominal wall is critical when planning incisions, raising myocutaneous flaps and reconstructing the abdominal wall during ventral hernia repair (Fig. 61.1).

Superior epigastric artery and veins

The superior epigastric artery is a terminal branch of the internal thoracic artery. It arises at the level of the sixth costal cartilage and descends between the costal and xiphoid slips of the diaphragm, accompanied by two or more veins that drain to the internal thoracic vein (see Fig. 61.1; Fig. 61.2). The vessels pass anterior to the lower fibres of transversus thoracis and the upper fibres of transversus abdominis before entering the rectus sheath, where they run inferiorly behind rectus abdominis. They anastomose with the inferior epigastric arteries, usually above the level of the umbilicus, in one of several potential branching patterns (Rozen et al 2008).

Branches supply rectus abdominis and perforate the anterior lamina of the rectus sheath to supply the abdominal skin. A branch given off in the upper rectus sheath passes anterior to the xiphoid process of the sternum and anastomoses with a corresponding contralateral branch. This vessel may give rise to bleeding during surgical incisions that extend up to and alongside the xiphoid process. The superior epigastric

artery also gives small branches to the anterior part of the diaphragm. On the right, small branches reach the falciform ligament, where they anastomose with branches from the hepatic artery.

Inferior epigastric artery and veins

The inferior epigastric artery (often referred to as the deep inferior epigastric artery in clinical practice in order to distinguish it from the superficial (inferior) epigastric artery) originates from the medial aspect of the external iliac artery just proximal to the inguinal ligament (see Figs 61.1–61.2; Fig. 61.3). Its accompanying veins, usually two, unite to form a single vein that drains into the external iliac vein (Rozen et al 2009). It curves forwards in the anterior extraperitoneal tissue and ascends obliquely along the medial margin of the deep inguinal ring. It lies posterior to the spermatic cord, separated from it by the transversalis fascia. It pierces the transversalis fascia and enters the rectus sheath by passing anterior to the arcuate line. In this part of its course, it is visible through the parietal peritoneum of the anterior abdominal wall and forms the lateral umbilical fold. Disruption of the artery at this site by surgical incisions (e.g. insertion of laparoscopic ports or abdominal drains) may result in a haematoma that can expand to considerable size because of the absence of adjacent tissue against which the bleeding can be tamponaded.

The vascular supply to the anterior abdominal wall can be divided into three zones. Zone I represents the epigastrium and central anterior abdominal wall within the region of rectus abdominis, and is supplied by the superior and inferior epigastric vessels. Zone II consists of the lower anterior abdominal wall below zone I and is predominantly supplied by the superficial epigastric, superficial external pudendal, and superficial circumflex iliac arteries. Zone III is lateral to zone I and is supplied by the musculophrenic, lower intercostal, subcostal and lumbar arteries.

The distance of the vertically running superior and deep inferior epigastric vessels from the midline is relevant to siting surgical incisions (Table 61.1).

Table 61.1 Distances of superior and deep inferior epigastric arteries from midline

Level	Left	Right
Xiphoid cartilage	4.5±0.1 cm	4.4±0.1 cm
Between xiphoid and umbilicus	5.4±0.2 cm	5.5±0.2 cm
Umbilicus	5.6±0.1 cm	5.9±0.1 cm
Between umbilicus and pubic symphysis	5.3±0.1 cm	5.3±0.1 cm
Pubic symphysis	7.5±0.1 cm	7.5±0.1 cm

Adapted from Saber AA, Meslemani AM, Davis R, Pimentel R 2004 Safety zones for anterior abdominal wall entry during laparoscopy: a CT scan mapping of epigastric vessels. Ann Surg 239:182–5.

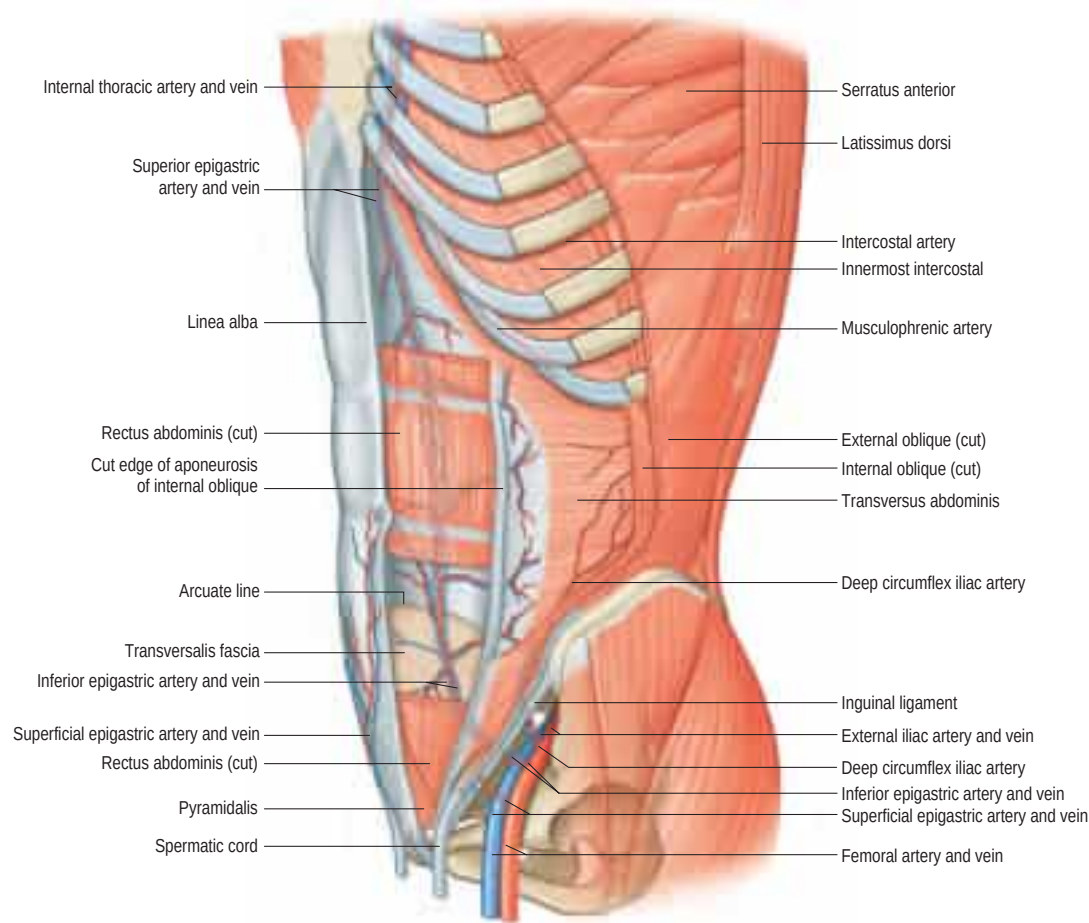


Fig. 61.2 The deep muscles and arterial supply of the anterolateral abdominal wall. The greater part of the left rectus abdominis has been removed to show the superior and inferior epigastric vessels.

The inferior epigastric arteries ascend and anastomose with their superior counterpart without branching in about 30% of cases (El-Mrakby and Milner 2002). Branching into two vessels before anastomosis is the most common pattern, accounting for almost 60% of cases, with a trifurcation being present in the remainder. The inferior epigastric arteries have an average diameter of approximately 3 mm at their origin, compared to an average diameter of 1.6 mm at the origin of the superior epigastric arteries, presumably explaining why the inferior epigastric arteries provide the 'dominant' supply to rectus abdominis. Preliminary ligation of the inferior epigastric artery is often performed when preparing a myocutaneous flap using the mid or lower rectus abdominis based on the superior epigastric artery; this encourages the augmentation of the superior epigastric arterial supply.

Branches of the inferior epigastric artery anastomose with branches of the superior epigastric artery within the rectus sheath posterior to rectus abdominis at a variable level above the umbilicus (Rozen et al 2008). Other branches anastomose with terminal branches of the lower five posterior intercostal, subcostal and lumbar arteries at the lateral border of the rectus sheath. Inferolaterally, branches anastomose with the deep circumflex iliac artery. The inferior epigastric artery ascends along the medial margin of the deep inguinal ring. The vas deferens in the male, or the round ligament in the female, passes medially after hooking around the artery at the deep inguinal ring. The artery forms the lateral border of Hesselbach's inguinal triangle, an important landmark in laparoscopic inguinal hernia repair; the inferior border of the triangle is formed by the inguinal ligament, and the medial border is formed by the lateral margin of rectus abdominis.

The inferior epigastric artery also gives off the cremasteric artery, a pubic branch, and muscular and cutaneous branches. The cremasteric artery accompanies the spermatic cord in males, supplies cremaster and the other coverings of the cord and anastomoses with the testicular artery. In females, the artery is small and accompanies the round ligament. A pubic branch, near the femoral ring, descends posterior to the pubis and anastomoses with the pubic branch of the obturator artery. The pubic branch of the inferior epigastric artery may be larger than the obturator artery and supply most of the obturator artery territory in the thigh, in which case it is referred to as the aberrant obturator artery (Pai et al 2009). It lies close to the medial border of the femoral ring and may be damaged in medial dissection of the ring during femoral or

laparoscopic inguinal hernia repair or with pelvic fractures. Muscular branches supply the abdominal muscles and peritoneum, and anastomose with the circumflex iliac and lumbar arteries. Cutaneous branches perforate the aponeurosis of external oblique, supply the skin and anastomose with branches of the superficial epigastric artery. These musculocutaneous perforators have been mapped in detail because they are particularly important to plastic surgeons undertaking reconstructive surgery with (myo)cutaneous flaps (Rozen et al 2008).

Occasionally, the inferior epigastric artery arises from the femoral artery. It then ascends anterior to the femoral vein to follow its usual abdominal course. Rarely, it arises from the external iliac artery in common with an aberrant obturator artery or from the obturator artery.

The superior and inferior epigastric arteries are important sources of collateral blood flow between the internal thoracic artery and the external iliac artery when aortic blood flow is compromised. Small tributaries of the inferior epigastric vein draining the skin around the umbilicus anastomose with terminal branches of the umbilical vein draining the umbilical region via the falciform ligament. These portosystemic anastomoses may open widely in cases of portal hypertension, when portal venous blood may drain into the systemic circulation via the inferior epigastric veins. The pattern of dilated superficial veins radiating from the umbilicus is referred to as the 'caput medusae'.

Posterior intercostal, subcostal and lumbar arteries

The tenth and eleventh posterior intercostal arteries, the subcostal artery, and the lumbar arteries pierce the posterior aponeurosis of transversus abdominis to enter the neurovascular plane of the abdominal wall deep to internal oblique (Fig. 61.4). The location of these arteries and their accompanying segmental nerves is of clinical importance when creating myofascial flaps during abdominal wall reconstruction. The arteries on either side run forwards, giving off muscular branches to the overlying internal and external oblique, before anastomosing with the lateral branches of the superior and inferior epigastric arteries at the lateral border of the rectus sheath (see Fig. 61.4). Perforating cutaneous vessels run vertically through the muscles to supply the overlying skin and subcutaneous tissue. A small contribution to the

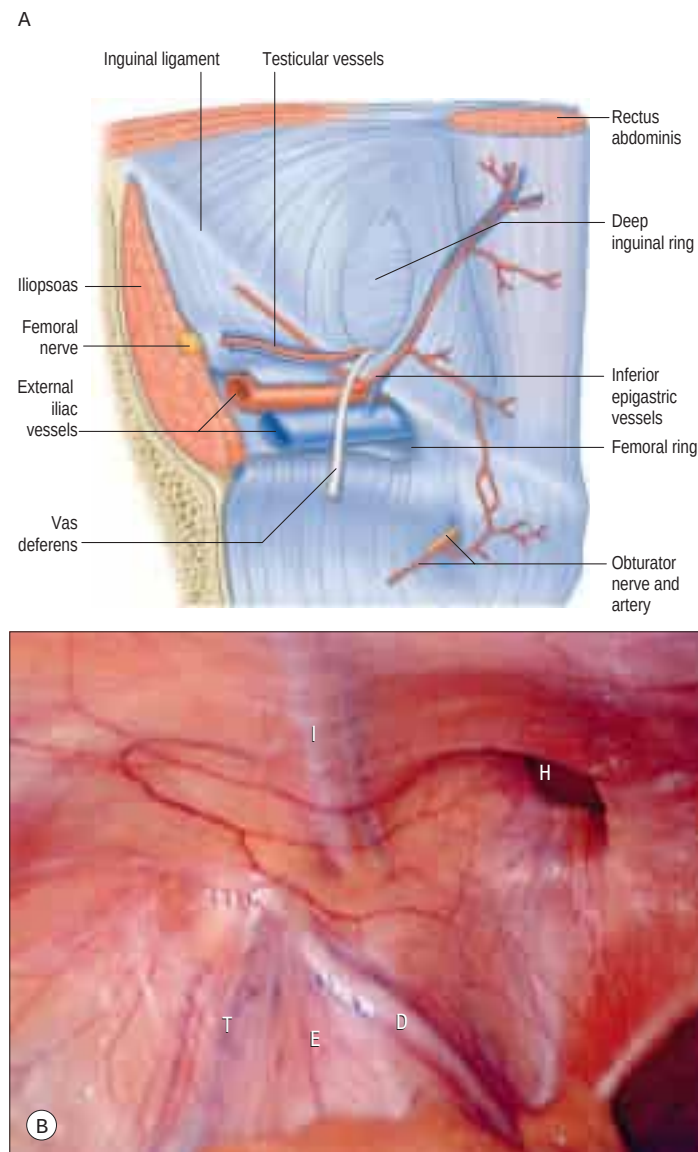


Fig. 61.3 **A**, The deep aspect of the lower part of the anterior abdominal wall of the left side. The femoral and deep inguinal rings are displayed, together with the vessels and other structures in relation to them and also the opening into the obturator canal. **B**, A laparoscopic view showing the parietal peritoneum covering the area. Abbreviations: D, vas (ductus) deferens; E, external iliac vessels; H, orifice of direct inguinal hernia; I, inferior epigastric vessels; T, testicular vessels. (B, With permission from Drake RL, Vogl AW, Mitchell A (eds), *Gray's Anatomy for Students*, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

supply of the lower abdominal muscles comes from branches of the deep circumflex iliac arteries.

The anterior abdominal wall is also supplied by branches of the femoral artery: namely, the superficial epigastric, superficial circumflex iliac, and superficial external pudendal arteries, and by the deep circumflex iliac artery arising from external iliac artery (see Fig. 78.7A).

Lymphatic drainage

Superficial lymphatic vessels accompany the subcutaneous blood vessels immediately below the dermis (Tourani et al 2013). Vessels from the lumbar and gluteal regions run with the superficial circumflex iliac vessels, and those from the infra-umbilical skin run with the superficial epigastric vessels. Both drain into superficial inguinal nodes. The supra-umbilical region is drained by vessels draining to axillary and parasternal nodes.

The deep lymphatic vessels accompany the deeper arteries. Laterally, they run either with the lumbar arteries to drain into the lateral aortic nodes, or with the intercostal and subcostal arteries to posterior mediastinal nodes. Lymphatics in the upper anterior abdominal wall run with the superior epigastric vessels to parasternal nodes while those in the

lower abdominal wall run with the deep circumflex iliac and inferior epigastric arteries to external iliac nodes.

SEGMENTAL NERVES

The ventral rami of the sixth to eleventh intercostal nerves, the subcostal nerve (twelfth thoracic) and first lumbar nerve (iliohypogastric and ilioinguinal nerves) supply the muscles and skin of the anterior abdominal wall (Rozen et al 2008). The seventh to the twelfth thoracic ventral rami continue anteriorly from the intercostal and subcostal spaces into the abdominal wall (Fig. 61.5). Approaching the costal margin, the seventh to tenth nerves curve medially across the deep surface of the costal cartilages between the digitations of the diaphragm and transversus abdominis. The subcostal nerve gives a branch to the first lumbar ventral ramus (dorsolumbar nerve) that contributes to the lumbar plexus (Ch. 62). It accompanies the subcostal vessels along the inferior border of the twelfth rib, passing behind the lateral arcuate ligament and kidney, and anterior to the upper part of the quadratus lumborum.

All these segmental nerves run anteriorly within a thin layer of fascia in the neurovascular plane between transversus abdominis and internal oblique, where they branch and interconnect with adjacent nerves (Rozen et al 2008). Muscular branches innervate transversus abdominis and internal and external oblique. Cutaneous branches supply the skin of the lateral and anterior abdominal walls. The thoracic nerves enter the rectus sheath at its lateral margin and pass posterior to rectus abdominis, where they again intercommunicate. Each nerve then pierces rectus abdominis from its posterior aspect and gives off muscular branches to this muscle (and a branch to pyramidalis from the subcostal nerve), and cutaneous branches that pierce the anterior rectus sheath to supply overlying skin.

The ninth intercostal nerve supplies skin above the umbilicus, the tenth supplies skin that consistently includes the umbilicus, and the eleventh supplies skin below the umbilicus (see Fig. 16.10; Fig. 61.5). The subcostal nerve supplies the anterior gluteal skin just below the iliac crest, and the skin of the lower abdomen and inguinal region (overlapping with the L1 dermatome in this region) (Lee et al 2008). A typical dermatome map of the anterior abdominal wall is shown in Figure 16.10. The ventral rami of the lower intercostal and subcostal nerves also provide sensory fibres to the costal parts of the diaphragm and parietal peritoneum.

The transversus abdominis plane (TAP) block is a regional anaesthetic technique for abdominal surgery. Using ultrasound imaging guidance, local anaesthetic is injected into the neurovascular plane between internal oblique and transversus abdominis, targeting the segmental nerves of the anterolateral abdominal wall.

Lesions of the intercostal nerves

The anterolateral abdominal wall muscles are innervated by several segmental nerves and injury to a single nerve does not produce a clinically detectable loss of muscle tone. The overlap between sequential dermatomes means that significant cutaneous anaesthesia is appreciated only after sectioning at least two sequential nerves.

MUSCLES

ANTEROLATERAL MUSCLES OF THE ABDOMEN

Rectus abdominis, pyramidalis, external oblique, internal oblique and transversus abdominis constitute the anterolateral muscles of the abdomen. They act together to perform a range of functions, some of which involve the generation of a positive pressure within one or more body cavity. Although many of these activities may occur with no 'forced assistance', expiration, defecation and micturition may be aided by the generation of a positive intra-abdominal pressure. Parturition, coughing and vomiting always require such a positive pressure. Under resting conditions, the tone developed within the muscles provides support for the abdominal viscera and retains the normal contour of the abdomen. The consequences of diminished muscular support can be seen in patients who have massive ventral hernias or conditions like 'prune belly syndrome', where there is congenital deficiency or absence of these muscles.

Active contraction of the muscles provides an important role in the maintenance of abdominal wall tone. The compression of the abdominal cavity required to increase intra-abdominal pressure is brought

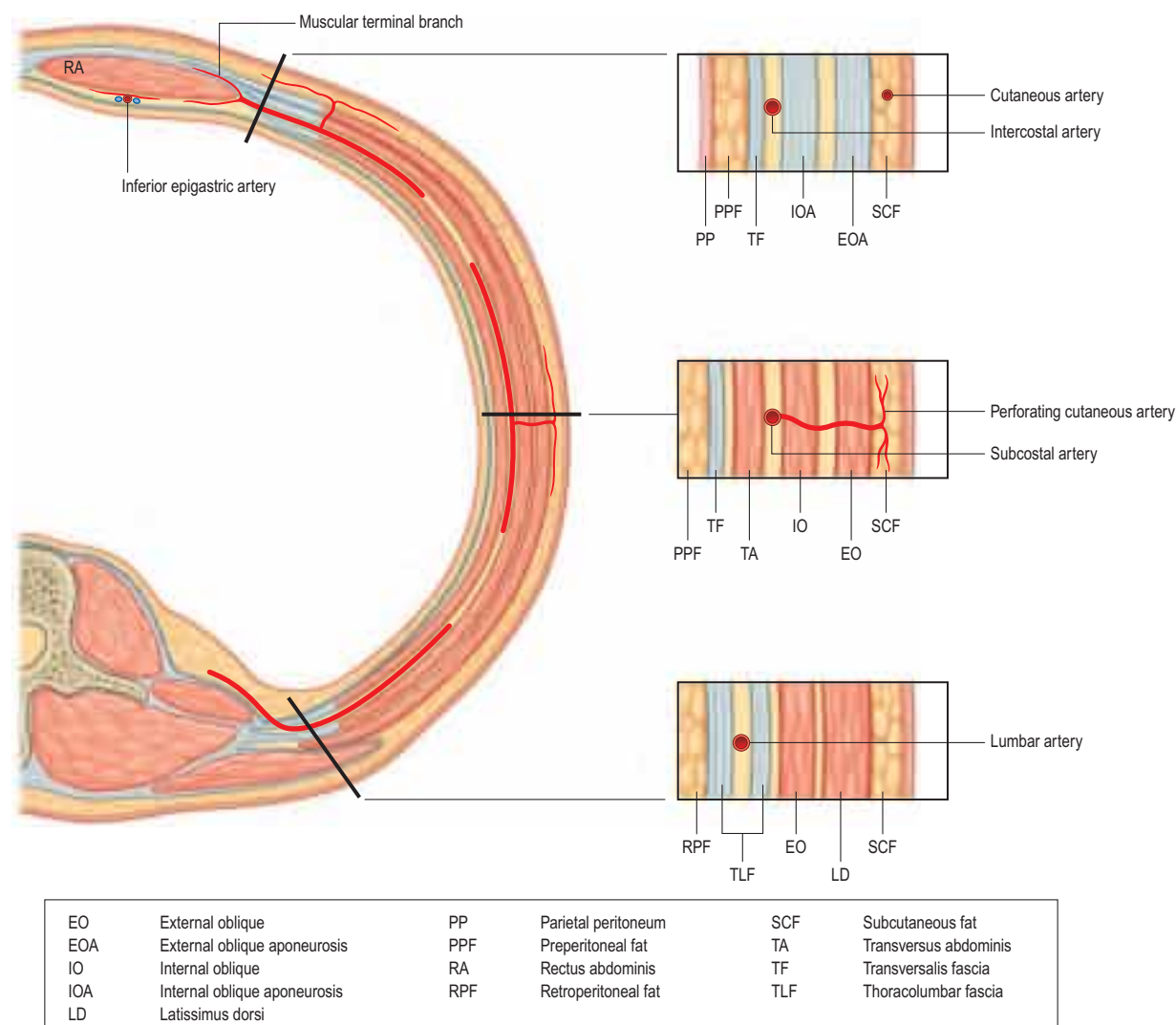


Fig. 61.4 The arrangement of the anterolateral abdominal wall vessels at the level of the mid-abdomen.

about mainly by contraction of the diaphragm. The bony pelvis, spine and lower thoracic cage provide rigidity to part of the abdominal wall. During the generation of positive intra-abdominal pressure, the abdominal muscles act to hold the abdominal wall in a relatively fixed position, rather than to generate pressure directly; because the majority of the abdominal wall is muscular, the anterolateral abdominal wall muscles must be synchronously contracted to prevent displacement of the viscera and the resultant loss of pressure. The oblique muscles, acting through their anterior aponeuroses and the rectus sheath, provide the majority of this tension, although transversus abdominis and rectus abdominis contribute.

The anterolateral abdominal muscles contribute little to the movements of the trunk during normal sitting and standing; these movements are controlled predominantly by the paravertebral and spinal muscles. However, movements of the trunk against resistance or when the individual is supine require the anterolateral abdominal muscles. Rectus abdominis is the most important in these situations, producing anterior flexion of the trunk. If the pelvic girdle is fixed, flexion of the thoracic girdle occurs. With a fixed thoracic cage, contraction of rectus abdominis causes the pelvis to tilt and lift. Lateral flexion and rotation of the trunk against resistance is provided by unilateral contraction of the oblique muscles.

Rectus abdominis

Rectus abdominis is a paired, long, strap-like muscle that extends along the entire length of the anterior abdominal wall on either side of the linea alba (Fig. 61.6). It is widest in the upper abdomen. The muscle fibres of rectus abdominis are partially interrupted by three fibrous bands or tendinous intersections, which pass transversely or obliquely across the muscle. One is usually situated at the level of the umbilicus, another opposite the free end of the xiphoid process and a third about midway between the other two. They are rarely full-thickness and usually extend only half-way through the anterior thickness of the

muscle, fusing with the fibres of the anterior lamina of the rectus sheath. Occasionally, one or two incomplete intersections are present below the umbilicus. The intersections may represent the myosepta delineating the myotomes that form the muscle.

The medial border of each rectus abdominis abuts the linea alba. Its lateral border may be visible on the surface of the anterior abdominal wall as a gently curved groove, the linea semilunaris, which extends from the tip of the ninth costal cartilage to the pubic tubercle. In a muscular individual it is readily visible, even when the muscle is not actively contracting, but in many normal and obese individuals it may be completely obscured.

Attachments Rectus abdominis arises by two tendons. The larger, lateral tendon is attached to the pubic crest and may extend beyond the pubic tubercle to the pectineal line. The medial tendon interlaces with the contralateral muscle and blends with ligamentous fibres covering the front of the pubic symphysis. Additional fibres may arise from the lower part of the linea alba. The pubic attachment of the tendon of rectus abdominis and its anterior sheath run over the anterior surface of the pubic symphysis and become continuous with the attachments of gracilis and adductor longus (Norton-Old et al 2013). Superiorly, rectus abdominis is attached by three slips of muscle to the fifth, sixth and seventh costal cartilages. The most lateral fibres are usually attached to the anterior end of the fifth rib; occasionally, this slip is absent or it extends to the fourth and third ribs. The most medial fibres are occasionally connected to the costoxiphoid ligaments and the side of the xiphoid process.

Vascular supply Rectus abdominis is supplied principally by the superior and inferior epigastric arteries, the latter being the dominant supply. Small terminal branches from the lower three posterior intercostal arteries, the subcostal artery, the lumbar arteries and the deep circumflex iliac artery may contribute, particularly at the lateral edges

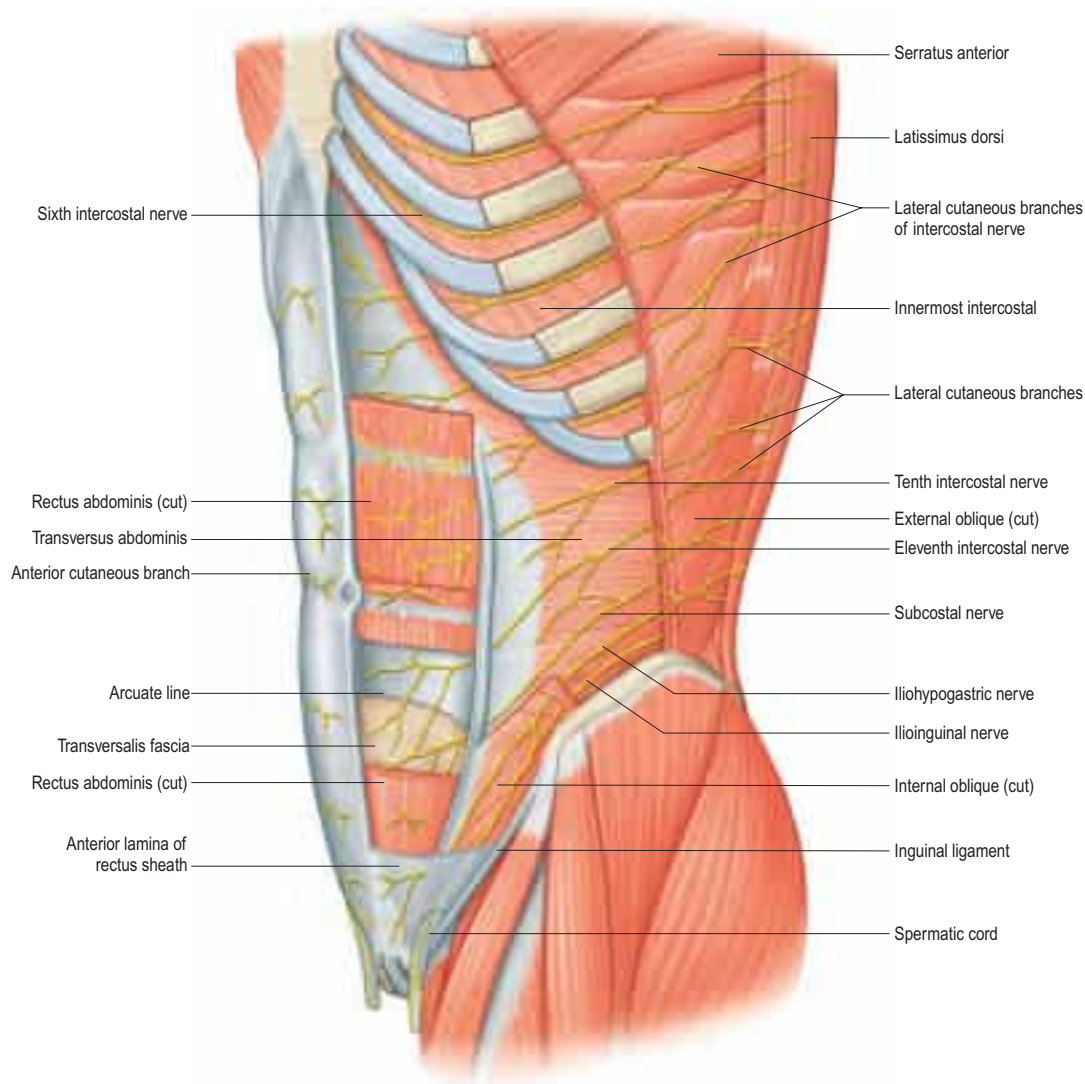


Fig. 61.5 The cutaneous branches of the lower intercostal and lumbar nerves. Portions of the muscles of the anterior abdominal wall have been removed, including most of the anterior layer of the rectus sheath and parts of rectus abdominis.

and the lower parts of the muscle, where they anastomose with small lateral branches of the epigastric arteries. Rectus abdominis provides a reliable and versatile myocutaneous flap, either pedicled or free, because of the excellent vascularity provided by the epigastric vessels and because the muscle belly can be separated relatively easily from its surrounding sheath. The upper half of the muscle may be used for breast reconstruction, and the lower half may be transposed to the groin and upper thigh or rotated on its lower attachments and delivered into the perineum for reconstruction after radical pelvic and perineal resections.

Innervation Rectus abdominis is innervated segmentally by the terminal branches of the ventral rami of the lower six or seven thoracic spinal nerves. It may also receive a branch from the ilioinguinal nerve (Rozen et al 2008).

Actions The recti contribute to flexion of the trunk and the maintenance of abdominal wall tone required during straining.

Rectus sheath

Rectus abdominis on each side is enclosed by a fibrous sheath (Figs 61.7–61.10). The anterior portion of this sheath extends the entire length of the muscle and fuses with periosteum and ligaments at sites of the muscle's attachments. The posterior part of the sheath is complete behind the upper two-thirds of the muscle but absent below this level, which corresponds to approximately one-third of the distance between the umbilicus and the pubis (Loukas et al 2008). The termination of the posterior rectus sheath is usually gradual but may be abrupt and marked by a clearly visible curved horizontal line known as the arcuate line (of Douglas). Below this level, rectus abdominis lies on the transversalis fascia and extraperitoneal connective tissue.

The rectus sheath is formed from the aponeuroses of all three lateral abdominal muscles: namely, external oblique, internal oblique and transversus abdominis. Each aponeurosis is bilaminar; the fibres

from all three anterior leaves run obliquely upwards, whereas the posterior leaves run obliquely downwards at right angles to the anterior leaves. Above the arcuate line, the anterior rectus sheath is composed of both leaves of the aponeurosis of external oblique and the anterior leaf of the aponeurosis of internal oblique fused together. The posterior rectus sheath is composed of the posterior leaf of the aponeurosis of internal oblique and both leaves of the aponeurosis of transversus abdominis. Thus, both the anterior and posterior layers of the rectus sheath consist of three layers of fibres with the middle layer running at right angles to the other two. At the midline, the anterior and posterior layers are closely approximated. Fibres from each layer decussate to the opposite side of the sheath, forming a continuous aponeurosis with the contralateral muscles. Fibres also decussate anteroposteriorly, crossing from the anterior sheath to the posterior sheath. The dense fibrous line caused by this decussation is called the linea alba. The external oblique, internal oblique and transversus abdominis muscles can therefore be regarded as digastric muscles with a central tendon comprising the linea alba (Rizk 1980). The decussating fibres at the linea alba can be used to identify the midline during surgical incisions. Below the arcuate line, all three aponeuroses from the oblique and transversus abdominis muscles pass into the anterior rectus sheath (see Fig. 61.7).

Linea alba and umbilicus

The linea alba is a tendinous raphe extending from the xiphoid process to the pubic symphysis and pubic crest. It lies between the two recti and is formed by the interlacing and decussating aponeurotic fibres of external oblique, internal oblique and transversus abdominis. At its lower end, the linea alba has two attachments to the pubis: superficial fibres are attached to the pubic symphysis, and deeper fibres form a triangular lamella that is attached behind rectus abdominis to the posterior surface of both pubic crests (adminiculum lineae albae). The fundiform ligament of the penis and pyramidalis are attached to the suprapubic part of the linea alba. The linea alba is visible externally in those who

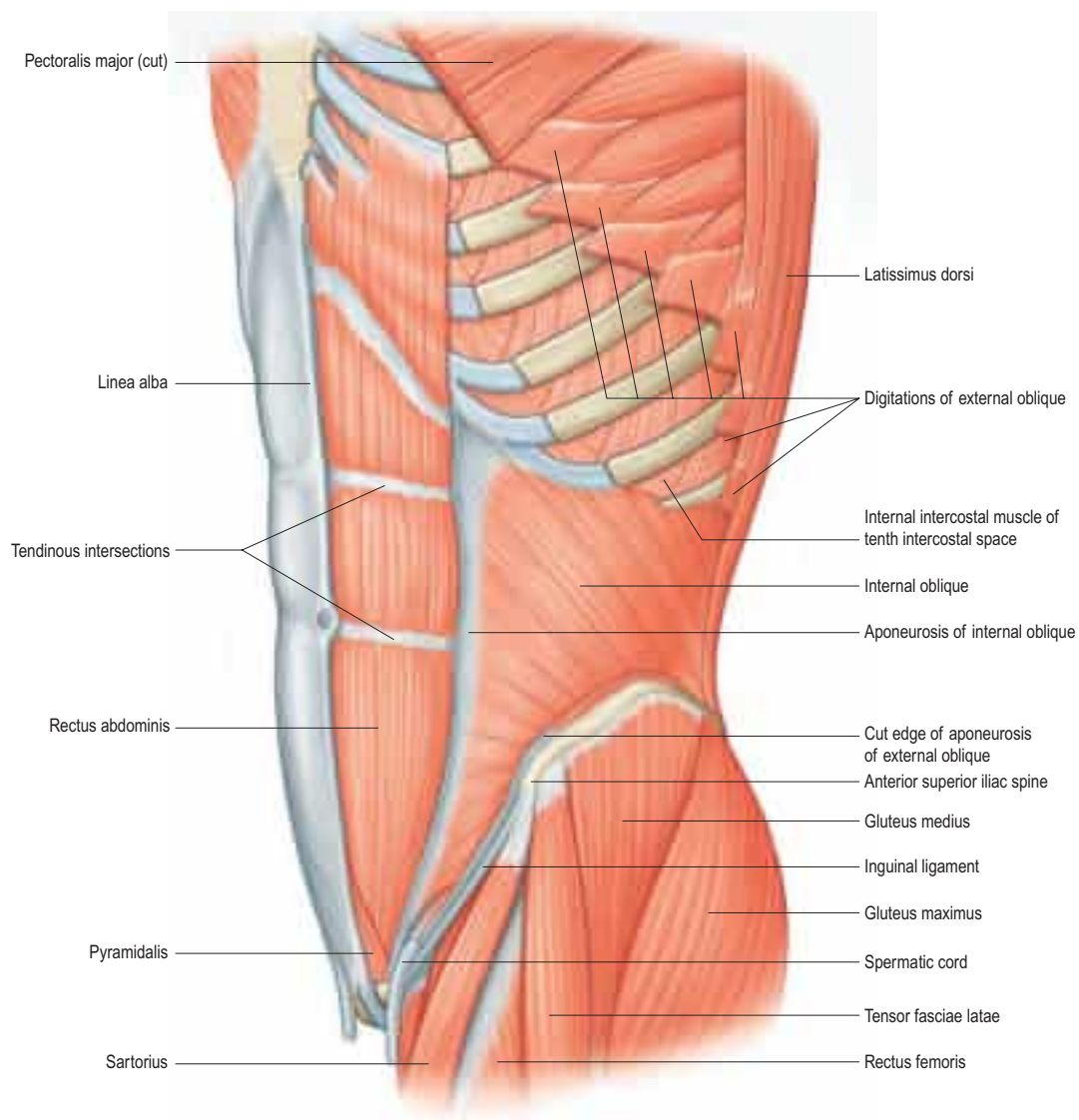


Fig. 61.6 Muscles of the left side of the trunk. External oblique has been removed to show internal oblique, but its digitations from the ribs have been preserved. The sheath of rectus abdominis has been opened and its anterior layer removed.

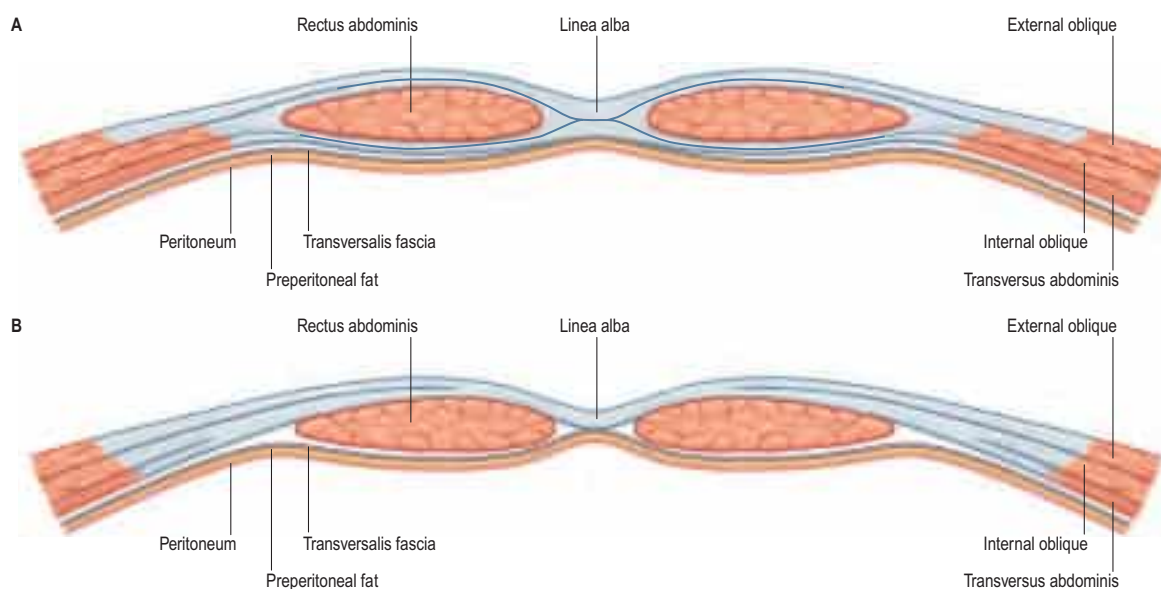


Fig. 61.7 Transverse sections through the anterior abdominal wall. **A**, Immediately above the umbilicus. **B**, Below the arcuate line. The bilaminar nature of each muscular aponeurosis is difficult to illustrate in cross-section. The fibres appear to fuse into a single sheet during formation of the rectus sheath. Note that rectus is supported directly by the transversalis fascia below the arcuate line.

are lean and muscular, as a shallow midline groove in the anterior abdominal wall. Its width varies along its length: it is wider above the umbilicus than below, and widest at the level of the umbilicus (Rath et al 1996). It is wider and thinner in women, in adults aged over 50 years, and in the obese and multiparous (Naraynsingh et al 2012). Its tensile strength is proportional to its thickness and density (Korenkov et al 2001). The linea alba is relatively bloodless but it is crossed superficially from side to side by a few small blood vessels.

The umbilicus is a fibrous cicatrix that lies a little below the mid-point of the linea alba, and is covered by an adherent area of skin. It

consists of skin, a fibrous layer (representing the area of fusion between the round ligament of the liver, the median umbilical ligament, and two medial umbilical ligaments), the transversalis fascia, the umbilical fascia surrounding the urachal remnant, and peritoneum (Fathi et al 2012). Its appearance and position are variable (Fathi et al 2012). In adults, it tends to lie at a relatively lower position with advancing age, in men, and in individuals with a higher body mass index.

In the fetus, the umbilicus transmits the umbilical vessels, urachus and, up to the third month of gestation, the vitelline or yolk stalk. It closes a few days after birth, but the vestiges of the vessels and urachus

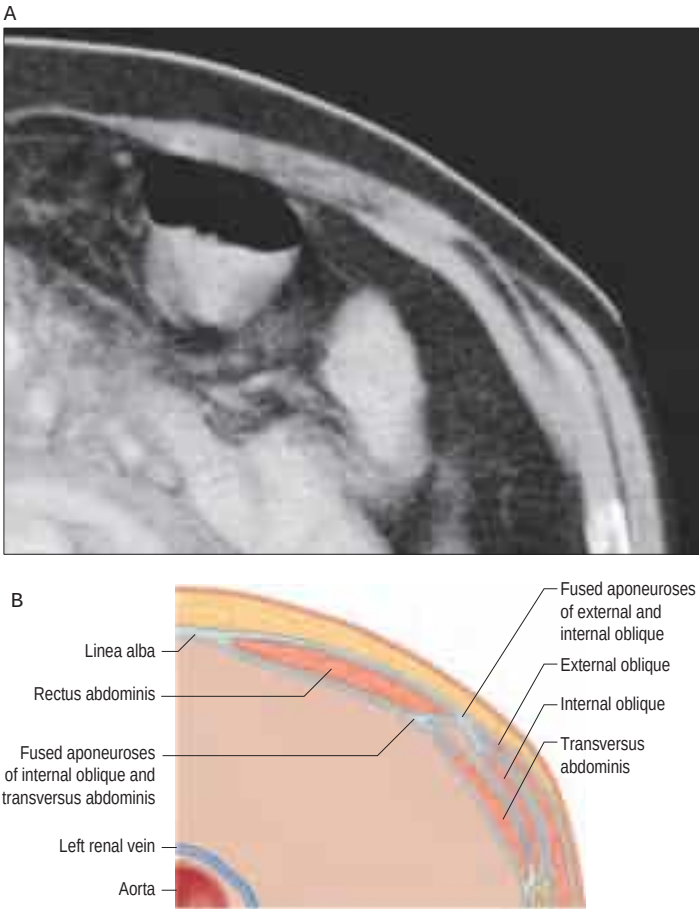


Fig. 61.8 The rectus sheath. Computed tomography scan (A) and diagram (B) of the anterior abdominal wall, demonstrating the formation of the rectus sheath above the umbilicus. The anterior rectus sheath is composed of both leaves of the aponeurosis of external oblique and the anterior leaf of the aponeurosis of internal oblique, fused together. The posterior rectus sheath is composed of the posterior leaf of the aponeurosis of internal oblique and both leaves of the aponeurosis of transversus abdominis.

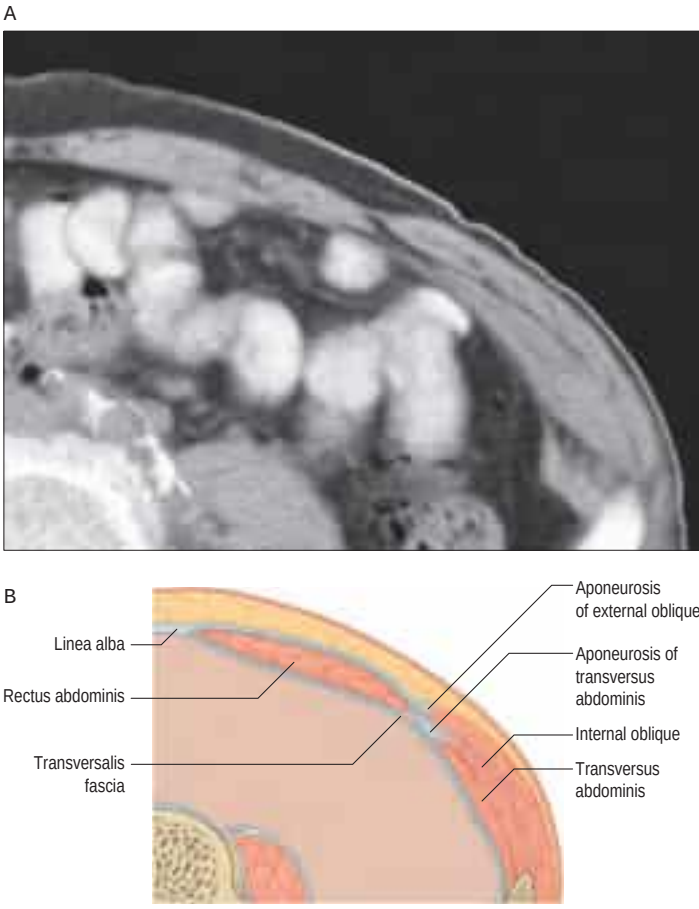


Fig. 61.9 The rectus sheath. Computed tomography scan (A) and diagram (B) of the anterior abdominal wall, demonstrating the formation of the rectus sheath below the umbilicus. The posterior rectus sheath is replaced by the transversalis fascia and extraperitoneal connective tissue.

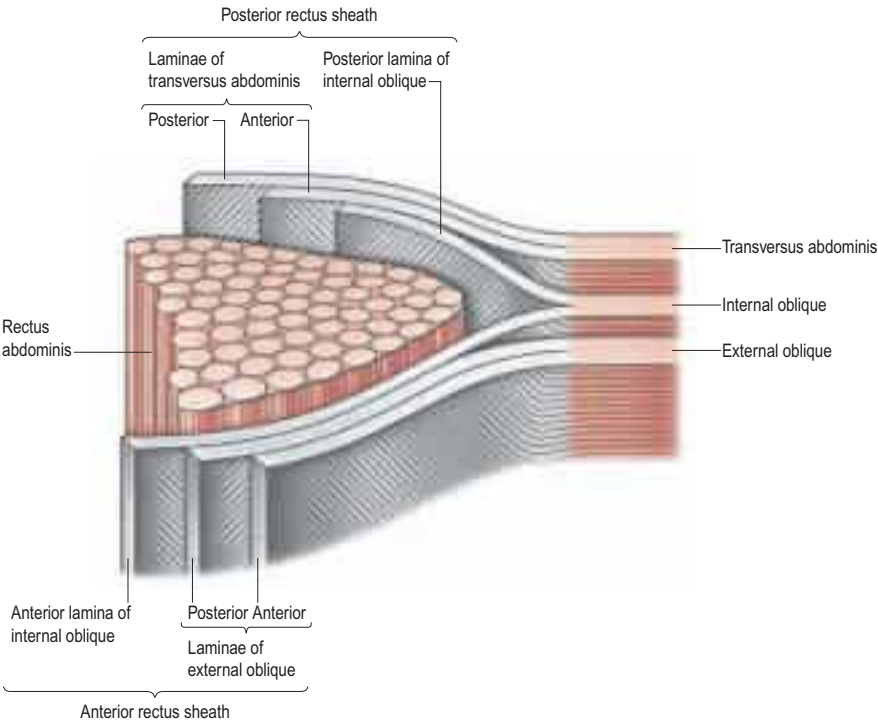


Fig. 61.10 The concept of bilaminar aponeuroses of the oblique and transversus abdominis muscles. Note that the fibres of the superficial and deep laminae are approximately at right angles; decussations occur as part of the linea alba.

remain attached to its deep surface. The remnant of the embryonic left umbilical vein forms the round ligament of the liver. The obliterated umbilical arteries form the medial umbilical ligaments on the under-surface of the anterior abdominal wall and are covered by the medial umbilical folds. The partially obliterated remains of the urachus persist as the median umbilical ligament and fold. Both congenital and acquired umbilical hernias are common; most childhood umbilical hernias close spontaneously and do not require surgical repair. The umbilicus is the most common site for laparoscopic access to the peritoneal cavity.

Divarication of the recti

Thinning and widening of the linea alba in the upper abdomen may occur, most commonly as a result of pregnancy, obesity or chronic straining (Akram and Matzen 2014). Abdominal viscera protrude beneath the thinned tissue as a broad midline bulge, particularly when intra-abdominal pressure is raised, and the recti become widely separated or divaricated. This state is not true herniation because all the layers of the abdominal wall in the region are intact.

Pyramidalis

Attachments Pyramidalis is a triangular muscle that lies in front of the lower part of rectus abdominis within the rectus sheath. It is attached by tendinous fibres to the anterosuperior margin of the pubis and to ligamentous fibres in front of the symphysis. The muscle diminishes in size as it runs upwards, and ends in a pointed apex that is attached medially to the linea alba. This attachment often lies midway between the umbilicus and pubis, but may occur higher. The muscle varies con-

siderably in size. It may be larger on one side than on the other, sometimes absent on one or both sides, or rarely doubled.

Vascular supply Pyramidalis is supplied by branches of the inferior epigastric artery. A small artery frequently crosses the midline posterior to the belly of the muscle to anastomose with the contralateral vessel.

Innervation Pyramidalis is usually supplied by the terminal branches of the subcostal nerve, the ventral ramus of T12, but it may be innervated wholly or in part by fibres from L1 travelling in the subcostal or ilioinguinal nerves (Tokita 2006).

Actions Pyramidalis contributes to tensing the lower linea alba but is of doubtful physiological significance.

External oblique

Attachments External oblique is the largest and most superficial of the three anterolateral abdominal muscles (Fig. 61.11). It curves around the lateral and anterior parts of the abdomen and is attached to the external surfaces and inferior borders of the lower eight ribs. The attachments rapidly become muscular and interdigitate with the lower fibres of serratus anterior and latissimus dorsi along an oblique line that extends downwards and backwards. The upper attachments are close to the cartilages of the corresponding ribs, the middle ones arise from the ribs at some distance from their cartilages, and the lowest are close to the apex of the cartilage of the twelfth rib. The fibres of external oblique diverge as they pass to their lower attachments. Those from the lower two ribs pass nearly vertically downwards and attach to the anterior half

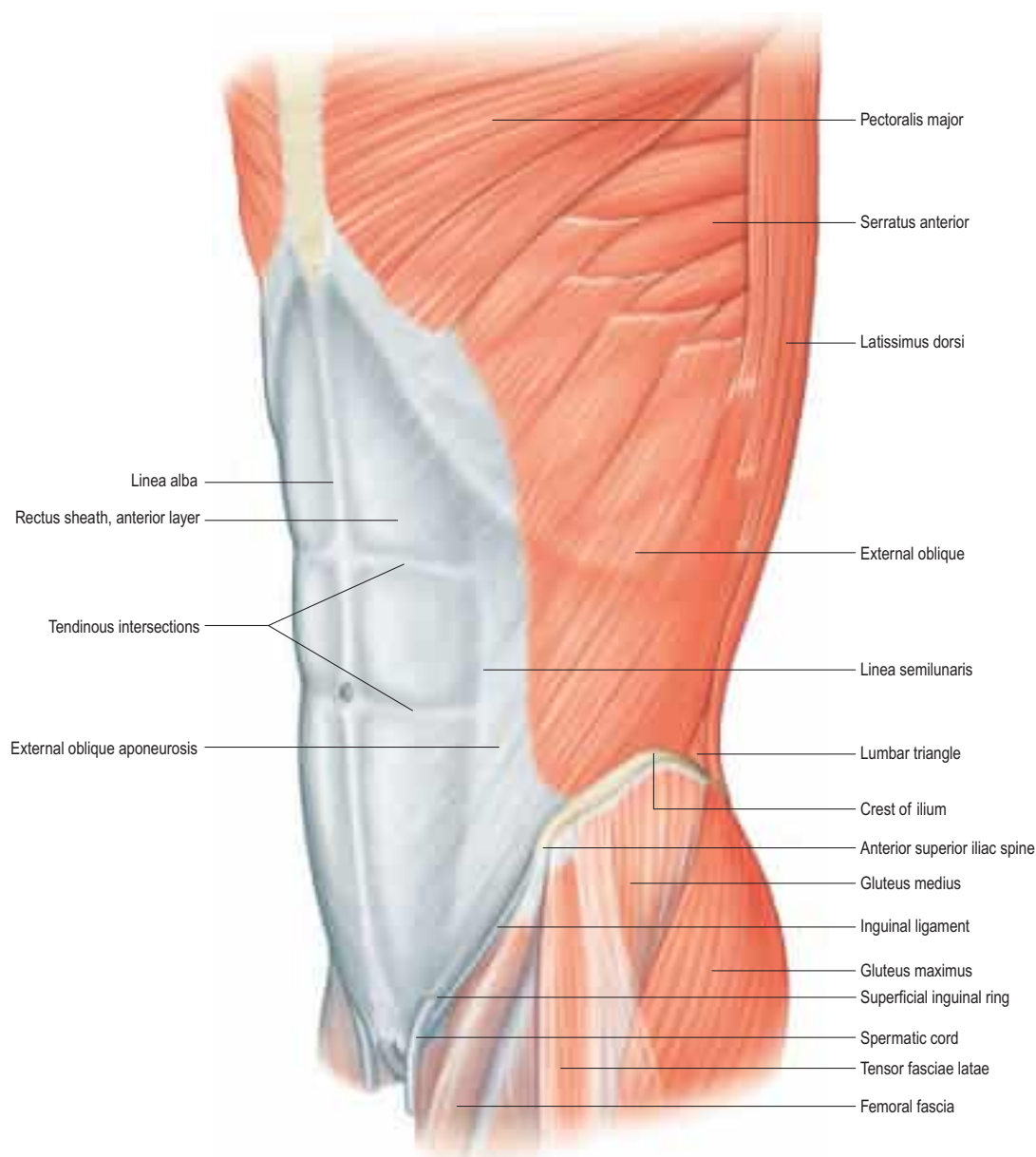


Fig. 61.11 The left anterolateral abdominal wall muscles.

or more of the outer lip of the iliac crest. The middle and upper fibres pass downwards and forwards, and end in the anterior aponeurosis, whose fibres cross the midline (see above). The junction between muscle and aponeurosis extends along a vertical line from the ninth costal cartilage to just below the level of the umbilicus; muscle fibres do not usually descend beyond a line from the anterior superior iliac spine to the umbilicus. External oblique has a free posterior border.

The inguinal ligament is formed by the inferior margin of the aponeurosis of external oblique extending between the anterior superior iliac spine and the pubic tubercle. The fibres of the aponeurosis of external oblique are not parallel to the long axis of the inguinal ligament; they approach the ligament obliquely at an angle of 10–20° and then turn medially to run along the ligament to reach the pubic tubercle (Lytle 1974). The deepest fibres of the aponeurosis spread out posteromedially to insert into the pectineal line.

The upper and lower rib attachments of the muscle may be absent. Digitations or even the entire muscle may be duplicated. The upper attachments of the muscle are sometimes continuous with pectoralis major or serratus anterior.

Vascular supply External oblique is mainly supplied by branches from the lower posterior intercostal and subcostal arteries above and the deep circumflex iliac artery below. There are additional smaller contributions (Schlenz et al 1999).

Innervation External oblique is innervated by the terminal branches of the lower five intercostal nerves and the subcostal nerve from the ventral rami of the lower six thoracic spinal nerves.

Actions External oblique contributes to the maintenance of abdominal tone, increasing intra-abdominal pressure, and lateral flexion of the trunk against resistance.

Inguinal ligament

The inguinal ligament is the thick lower border of the aponeurosis of external oblique that stretches between the anterior superior iliac spine and the pubic tubercle. Its medial half is curled in on itself, forming the gutter-like 'floor' of the inguinal canal. The ligament is not exactly linear but has an inferior and an anterior convexity. At its lower border, it is fused with the fascia lata. Laterally, it is fused with the iliopsoas fascia (Lytle 1974). At the medial end of the inguinal ligament, near its site of attachment to the pubic tubercle, some of its fibres extend posteriorly and laterally to attach to the pectineal line, forming the triangular, shelf-like lacunar ligament. Other fibres pass upwards and medially behind the superficial inguinal ring and external oblique aponeurosis to join the rectus sheath and linea alba; these constitute the reflected part of the inguinal ligament (Tubbs et al 2009). Fibres from both sides decussate in the linea alba.

Internal oblique

Attachments Internal oblique lies deep to external oblique for most of its course (see Fig. 61.6). It is thinner and less bulky than external oblique. Its fibres are traditionally stated to arise from the lateral two-thirds of the inguinal ligament but, in fact, arise from the corresponding length of a slightly deeper structure known as the iliopectineal arch (Acland 2008), a thickened band of iliopsoas fascia that passes downwards and medially from the anterior superior iliac spine to the iliopectineal eminence of the hip bone. Further laterally, internal oblique is attached to the anterior two-thirds of the iliac crest deep to the attachment of external oblique. Posteriorly, some fibres are attached to the thoracolumbar fascia. The fibres originating from the posterior end of the iliac attachment pass obliquely upwards and are attached to the inferior borders and tips of the lower three or four ribs and their cartilages. Here, the attachments merge with those of the internal intercostal muscles. The uppermost fibres form a short, free superior border. The fibres from the anterior end of the iliac crest diverge and end in the anterior aponeurosis, which gradually broadens from below upwards. The uppermost part of the aponeurosis is attached to the cartilages of the seventh, eighth and ninth ribs. The fibres that originate adjacent to the inguinal ligament arch downwards and medially across the spermatic cord in the male and the round ligament of the uterus in the female. They become tendinous, fuse with the corresponding part of the aponeurosis of transversus abdominis, and attach to the pubic crest and medial part of the pectineal line, forming the conjoint tendon.

Vascular supply Internal oblique is mainly supplied by branches from the lower posterior intercostal and subcostal arteries, the inferior epigastric artery, and the deep circumflex iliac artery. There are several smaller contributions (Yang et al 2003).

Innervation Internal oblique is innervated by the terminal branches of the lower five intercostal nerves and the subcostal nerve from the ventral rami of the lower six thoracic spinal nerves. In addition, it receives a small contribution from the iliohypogastric and ilioinguinal nerves derived from the ventral ramus of the first lumbar spinal nerve.

Actions Internal oblique contributes to the maintenance of abdominal tone, increasing intra-abdominal pressure, and lateral flexion of the trunk against resistance.

Transversus abdominis

Attachments Transversus abdominis is the deepest of the anterolateral abdominal muscles (Fig. 61.12). It is attached to the iliopectineal arch deep to the lateral third of the inguinal ligament, the anterior two-thirds of the inner lip of the anterior segment of the iliac crest, the thoracolumbar fascia between the iliac crest and the twelfth rib, and the internal aspects of the lower six costal cartilages. The costal attachments interdigitate with those of the diaphragm. The muscle ends anteriorly in an aponeurosis; the lower fibres of the aponeurosis curve downwards and medially, together with those of the aponeurosis of internal oblique, and insert into the pubic crest and pectineal line to form the conjoint tendon. A few muscle fibres may run from the lower border of transversus abdominis to the inguinal ligament and reinforce the interfoveolar ligament (see above). The remainder of the aponeurosis passes medially and the fibres decussate at, and blend with, the linea alba. The upper costal and anterior iliac fibres of transversus abdominis are short and the thoracolumbar fibres are the longest. Near the xiphoid process, the aponeurosis is formed only a few centimetres from the linea alba and so the muscular part of transversus abdominis extends behind rectus abdominis into the posterior layer of the rectus sheath. The medial edge of the muscle, at the origin of the aponeurosis, curves downwards and laterally, and is furthest from the lateral edge of the rectus sheath at the level of the umbilicus. It then curves downwards and medially towards the superficial inguinal ring.

Occasional defects may occur in the lower muscular and aponeurotic parts of both internal oblique and transversus abdominis. The two muscles are sometimes fused and, rarely, transversus abdominis may be absent.

Vascular supply Transversus abdominis is supplied by branches from the lower posterior intercostal and subcostal arteries, the superior and inferior epigastric arteries, the superficial and deep circumflex iliac arteries and the posterior lumbar arteries.

Innervation Transversus abdominis is innervated by the terminal branches of the lower five intercostal nerves, the subcostal nerve and the iliohypogastric and ilioinguinal nerves. These arise from the ventral rami of the lower six thoracic and first lumbar spinal nerves.

Actions Transversus abdominis contributes mainly to the maintenance of abdominal tone and increasing intra-abdominal pressure.

Conjoint tendon

The conjoint tendon is formed from the lower fibres of internal oblique and the lower part of the aponeurosis of transversus abdominis. It is attached to the pubic crest and extends to a variable extent along the pectineal line (see Fig. 61.16). It descends behind the superficial inguinal ring and acts to strengthen the medial portion of the posterior wall of the inguinal canal. Medially, the upper fibres of the tendon fuse with the anterior wall of the rectus sheath, and laterally, some fibres may blend with the interfoveolar ligament.

Cremaster

Attachments Cremaster consists of loosely arranged muscle fasciculi lying along the spermatic cord or round ligament of the uterus. It is variable in thickness but thickest in young men. Together with connective tissue, it forms an incomplete coating around the spermatic cord, known as the cremasteric fascia, which extends down around the testis deep to the external spermatic fascia. The muscle consists of a lateral part that arises mainly from the inferomedial border of internal oblique and transversus abdominis but also has attachments to the middle of the inguinal ligament, and a medial part attached to the pubic tubercle, lateral pubic crest and the lower border of transversus abdominis; this part is variably developed, and may be absent (Shafik 1977). The longitudinal muscle fibres spread out over the spermatic cord as it approaches the superficial inguinal ring. Cremaster is composed of both striated and smooth muscle fibre bundles (Kayalioglu et al 2008).

Vascular supply Cremaster is supplied by the cremasteric artery, a branch of the inferior epigastric artery.

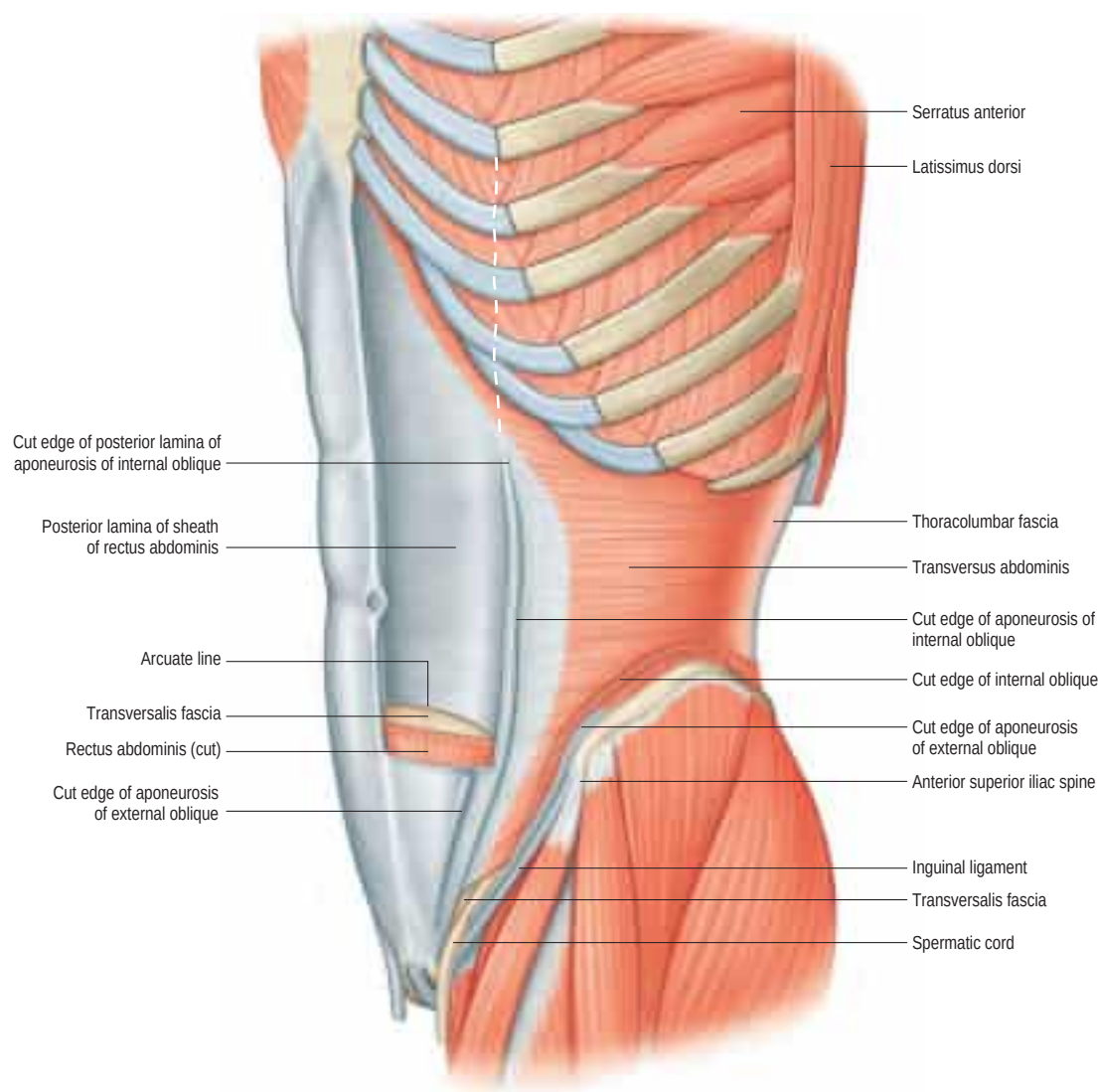


Fig. 61.12 The left transversus abdominis. The aponeurosis of transversus abdominis fuses into the posterior layer of the rectus sheath above the arcuate line. The position of the lateral border of rectus abdominis is shown by the dotted white line.

Innervation Cremaster is innervated by the genital branch of the genitofemoral nerve, derived from the first and second lumbar spinal nerves.

Actions Cremaster pulls the testis up towards the superficial inguinal ring. Although it contains striated muscle fibres, it is not usually under voluntary control. Stroking the skin of the medial side of the thigh evokes a reflex contraction of the muscle, the cremasteric reflex, which is most pronounced in boys. It may represent a protective reflex. The cremaster may also have a role in testicular thermoregulation since it is activated by the cold.

Inguinal canal

The inguinal canal is a natural passageway between the muscle layers of the anterior abdominal wall in the region of the groin (Figs 61.13–61.15; see Figs 76.15, 76.16). Its size and form vary with age, and although it is present in both sexes, it is best developed in the male. The canal is an oblique tunnel, with deep (internal) and superficial (external) openings or rings. It transmits the spermatic cord in males, the round ligament of the uterus in females, and the ilioinguinal nerve in both sexes.

Superficial inguinal ring

The superficial inguinal ring is a hiatus in the aponeurosis of external oblique, just above and lateral to the crest of the pubis. The ring is actually triangular, with its apex pointing laterally towards the anterior superior iliac spine. Although it varies in size, it does not usually extend laterally beyond the medial third of the inguinal ligament. The ring is smaller in the female. The base of the triangular opening lies along the crest of the pubis. Its sides are the lateral and medial crura of the opening in the aponeurosis. The lateral crus is stronger and is attached to the pubic tubercle. The medial crus is thinner and its fibres attach to the front of the pubic symphysis and interlace with those from the opposite side. A few intercrural fibres arch above the apex of the superficial inguinal ring (see Fig. 61.13). Some fibres from the external

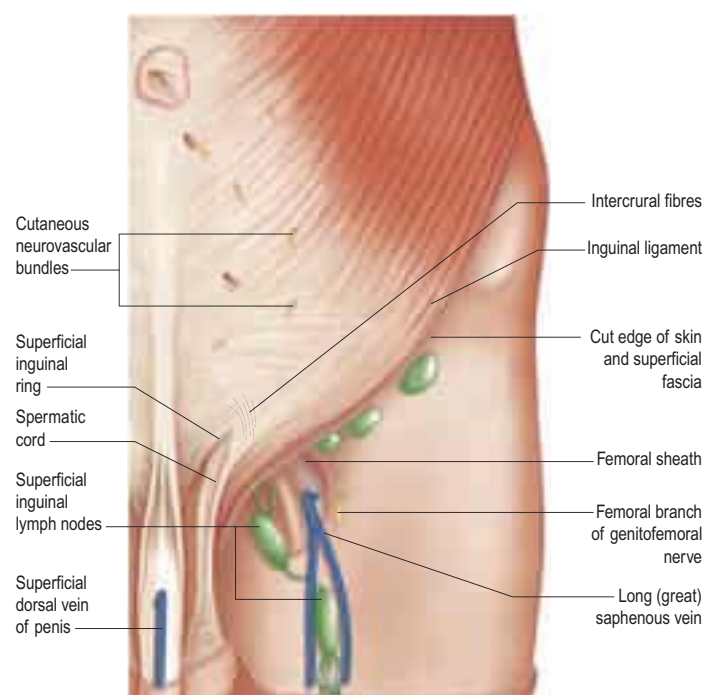


Fig. 61.13 Superficial structures of the inguinal region and lower part of the anterior abdominal wall on the left side.

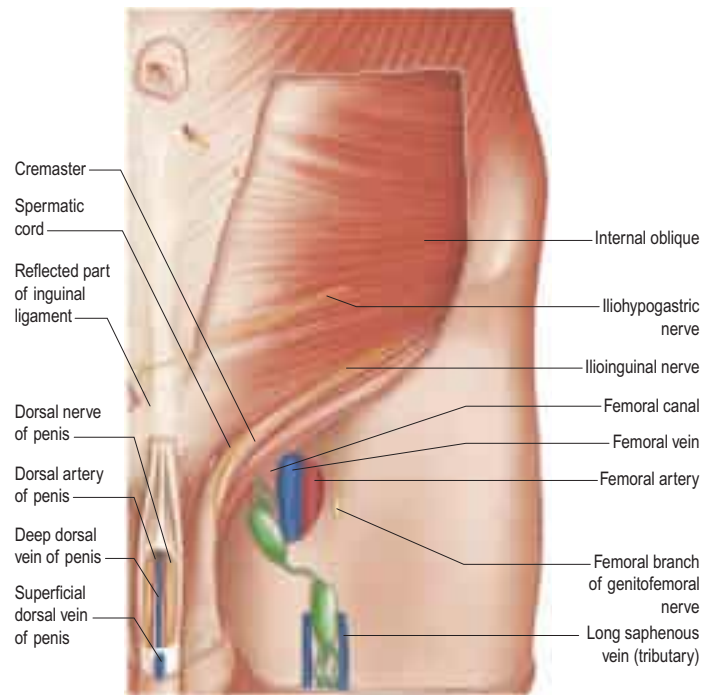


Fig. 61.14 Dissection of the regions shown in Figure 61.13, with part of external oblique removed.

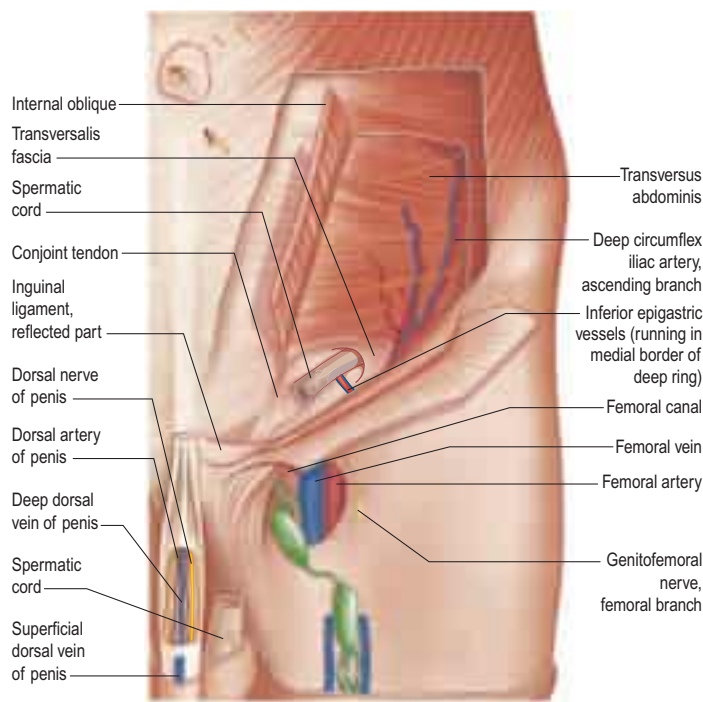


Fig. 61.15 Dissection of the inguinal region and lower part of the anterior abdominal wall on the left side, with parts of external and internal oblique muscles removed.

oblique aponeurosis and its overlying fascia continue downwards from the crura of the ring over the spermatic cord, and form the delicate external spermatic fascia; consequently, the 'ring' appears less of a discrete opening in the living.

Deep inguinal ring

The deep inguinal ring is an opening in the transversalis fascia, approximately midway between the anterior superior iliac spine and the pubic symphysis, and about 1 cm above the inguinal ligament (Hale et al 2010). It is oval, with a roughly vertical long axis. Its size varies between individuals but it is usually 1–2 cm wide in adults and larger in the male. It is related above to the arched lower margin of transversus abdominis and medially to the interfoveolar ligament (see Fig. 61.16). The inferior epigastric vessels run in the medial border of the deep inguinal ring. Traction on the fascial ring exerted by contraction of internal oblique may narrow the opening when intra-abdominal pressure is increased.

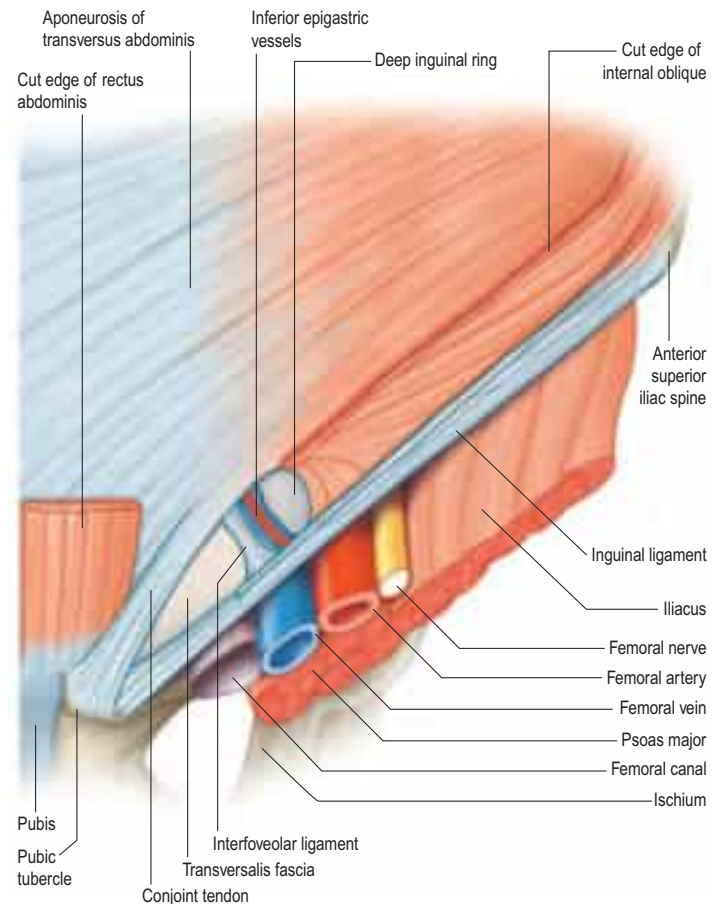


Fig. 61.16 Deep structures of the inguinal canal. The aponeurosis of external oblique has been removed. For clarity, the fibres of internal oblique and rectus abdominis have been divided, and the structures passing posteroinferiorly to the inguinal ligament have been excluded.

Boundaries

The inguinal canal slants obliquely downwards and medially, parallel to and just above the medial part of the inguinal ligament. It extends from the deep to the superficial inguinal rings; the length depends on the age of the individual, but in the adult is between 3 and 6 cm long. The canal is bounded anteriorly by skin, superficial fascia and the aponeurosis of external oblique. In its lateral third, the anterior wall is reinforced by the muscular fibres of the internal oblique just above their origin from the iliopectineal arch. Posteriorly lie the reflected inguinal ligament, the conjoint tendon and the transversalis fascia, which separate it from extraperitoneal connective tissue and peritoneum. Superiorly lie the arched fibres of internal oblique and transversus abdominis, forming the conjoint tendon medially. Inferiorly is the union of the transversalis fascia with the inguinal ligament and, at the medial end, the lacunar ligament.

In the newborn, the deep and superficial rings are nearly superimposed and the canal is extremely short. In infants undergoing inguinal hernia repair, the canal is only about 1 cm long (Parnis et al 1997). As a child grows, the anterior abdominal wall muscles develop further, causing the positions of the rings to separate and the canal to lengthen. The canal becomes progressively more oblique so that, by adulthood, it has separate anterior and posterior walls, creating a 'shutter' effect. At the superficial ring and medial end of the anterior wall, where the canal is weakest, the posterior wall is strengthened by the conjoint tendon and reflected inguinal ligament. Increases in intra-abdominal pressure transmitted to the posterior wall of the canal are resisted by contraction of the three anterolateral abdominal wall muscles. The fibres of internal oblique and transversus abdominis, which form the conjoint tendon, are constantly active in standing; this activity increases during episodes of increased intra-abdominal pressure.

Relations

The inferior epigastric vessels are important medial relations of the deep inguinal ring (Fig. 61.16). They lie on the transversalis fascia as they ascend obliquely behind the conjoint tendon to enter the rectus sheath. The inguinal triangle (of Hesselbach) is an important clinical landmark

related to the posterior wall of the inguinal canal and is best appreciated by examining the inguinal canal region from within the abdomen (see Fig. 61.3). The triangle is bounded inferiorly by the medial third of the inguinal ligament, medially by the lower lateral border of rectus abdominis, and laterally by the inferior epigastric vessels.

Lacunar ligament

The lacunar ligament is a triangular band of fibrous tissue lying mainly posterior to the medial end of the inguinal ligament. It measures approximately 2 cm from base to apex and is a little larger in the male. It is formed from fibres of the medial end of the inguinal ligament together with fibres from the fascia lata of the thigh, which join the medial end of the inguinal ligament from below (Lytle 1974). The inguinal fibres run posteriorly and laterally to the medial end of the pectineal line and are continuous with the pectineal fascia. They form a near-horizontal, triangular sheet with a curved lateral border, which forms the medial border of the femoral canal. The apex of the triangle is attached to the pubic tubercle. A strong fibrous band, the pectineal ligament, extends laterally along the pectineal line from the pectineal attachment of the lacunar ligament (Faure et al 2001).

Myofascial flaps and component separation



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HERNIAS OF THE ANTERIOR ABDOMINAL WALL

A wide range of anterior abdominal wall hernias are described. These include inguinal, umbilical, incisional, para-umbilical, femoral, and rarer types such as Spigelian hernias (Larson and Farley 2002; Dabbas et al 2011).

INGUINAL HERNIA

Although the inguinal canal is arranged such that the weaknesses in the anterior abdominal wall caused by the deep and superficial inguinal rings are supported, the region is a common site of herniation, particularly in males. An inguinal hernia involves the protrusion of a viscus through the tissues of the inguinal region of the abdominal wall.

Indirect inguinal hernia

An indirect inguinal hernia arises through the deep inguinal ring lateral to the inferior epigastric vessels. Many indirect hernias are related to the abnormal persistence of a patent processus vaginalis, a tube-like extension of peritoneum through the inguinal canal that is present during normal development and normally becomes occluded after birth (p. 1215). Others are acquired as a result of progressive weakening of the posterior wall of the inguinal canal in the region of the deep inguinal ring. The hernia may pass through the deep ring or may expand the deep ring such that it is no longer a clear entity. Small indirect hernias lie below and lateral to the fibres of the conjoint tendon, but larger hernias often distort and thin the tendon superiorly. Small indirect hernias that do not protrude beyond the inguinal canal are covered by the same inner layers as the spermatic cord: namely, the internal spermatic fascia and cremaster. If the hernia extends through the superficial inguinal ring, it is also covered by external spermatic fascia. In hernias related to a persistent fully patent processus vaginalis, the hernia contents may descend as far as the tunica vaginalis anterior to the testis. In many individuals with a partial or fully patent processus vaginalis, an indirect hernia will manifest in childhood, but in others, an actual hernia into the potential sac may not develop until adult life, often as a consequence of increased intra-abdominal pressure or sudden muscular strain.

Direct inguinal hernia

A direct inguinal hernia arises medial to the inferior epigastric vessels. Direct hernias are always caused by an acquired weakness of the posterior wall of the inguinal canal (see Fig. 61.3B); as they enlarge, they frequently extend through the anterior wall of the inguinal canal or superficial inguinal ring, becoming covered by external spermatic fascia in the process. A direct inguinal hernia may closely resemble an indirect hernia and can be difficult to distinguish on clinical examination.

Clinical features of inguinal hernias

Indirect inguinal hernias often descend from lateral to medial, following the path of the inguinal canal, whereas direct inguinal hernias tend to protrude more directly anteriorly. With the hernia reduced, pressure applied over the region of the deep inguinal ring may prevent the appearance of an indirect hernia on standing or straining, but distinguishing an indirect from a direct inguinal hernia by clinical examination alone is not reliable (Ralphs et al 1980, Tromp et al 2014). Direct hernias are more likely to have a wide neck, making strangulation less likely.

UMBILICAL HERNIA

The most extreme variety of umbilical hernia is known as an omphalocele or exomphalos, a congenital malformation in which abdominal viscera, covered by a membrane, protrude through a wide umbilical defect (Fig. 61.17). The defect arises from a failure of closure of the umbilical ring after return of the herniated midgut loop in the embryo. The most common variety of umbilical hernia is caused by a weakness of the umbilical scar tissue, and is often seen in babies, especially those of African descent. The vast majority of these will close spontaneously during early childhood. Most umbilical hernias in adults are acquired as a result of stretching of the supporting umbilical fascia and are due to obesity and chronically increased intra-abdominal pressure (e.g. from multiple pregnancies or ascites).



FEMORAL HERNIA

The femoral sheath is continuous with the transversalis fascia anteriorly and the iliopsoas fascia posteriorly (p. 1338). The medial compartment of the sheath is the femoral canal, which typically contains lymphatics embedded in loose adipose connective tissue. The opening of the canal is the femoral ring, bounded anteriorly by the inguinal ligament, posteriorly by the pectineal ligament, medially by the crescentic lateral margin of the lacunar ligament and laterally by the femoral vein. A femoral hernia protrudes through the femoral ring, which is normally closed by a femoral septum of extraperitoneal tissue, and is therefore a site of potential weakness. In females, the ring is relatively wider and changes during pregnancy, which helps to explain why femoral hernias account for about 20% of all groin hernias in women but less than 1% of groin hernias in men (Whalen et al 2011). The ring also widens with advancing age. Variations in the attachment of the pectineal part of the lacunar ligament may also be a factor.

When a tongue of omentum or a loop of intestine bulges through the ring, it pushes out a hernial sac of peritoneum that is covered by extraperitoneal tissue (the femoral septum) and descends within the femoral canal to the saphenous opening. It is prevented from descending further by the configuration of the femoral sheath and by the attachment of fascia to the inferior rim of the saphenous opening. The hernial sac hence turns forwards, stretching the cribriform fascia and curving upwards over the inguinal ligament within the subcutaneous tissues. While in the canal, the hernia is usually small because it is contained by the surrounding tissues, but it enlarges as it expands into the subcutaneous tissues.

The small size of some femoral hernias explains why they can be easily missed on examination, particularly in the obese. They are also much more prone to strangulation than inguinal hernias (Whalen et al 2011). The site of strangulation varies: it may be at the neck of the hernial sac, or at the saphenous opening, but is most often at the junction of the falciform margin of the saphenous opening with the free edge of the pectineal part of the inguinal ligament. When the lacunar ligament is being divided to release the neck of the hernia, care must be taken to avoid or control an aberrant obturator artery (see above).

The pubic tubercle can be a useful landmark when attempting to distinguish an inguinal from a femoral hernia; a femoral hernia lies below and lateral to this landmark, whilst an inguinal hernia is above. However, clinical differentiation of groin hernias can be relatively unreliable (Hair et al 2001), prompting the use of imaging in suspected cases (Whalen et al 2011).

SPIGELIAN HERNIA

A Spigelian hernia is a protrusion of preperitoneal fat or a peritoneal sac through a congenital or acquired defect in the abdominal wall in

The goal of repairing massive ventral hernias lies in reapproximation of the rectus sheath, thereby restoring anatomy and maximizing the physiological potential of the abdominal wall. However, large fascial defects often will not permit primary reapproximation without undue tension. Many techniques have been developed that utilize the layered nature of the abdominal wall musculature, allowing additional mobilization of the fascial edges by 'releasing' incisions in myofascial layers (Rosen 2011). This section reviews some of the more common techniques. These repairs are often reinforced with synthetic or bioprosthetic mesh applied in various ways: onlay (superficial to the fascia), sublay (within the myofascial layers) or underlay (beneath the fascia and exposed to intraperitoneal contents).

The Ramirez technique, originally described in 1990, refers to bilateral release of the external oblique muscles, which can potentially provide up to 10 cm, 20 cm and 8 cm of medial mobilization at the upper, mid and lower abdomen, respectively (Ramirez et al 1990). A disadvantage of this technique is that, in order to access and isolate the external oblique muscles lateral to the linea semilunaris from a midline incision, large subcutaneous flaps must be raised. This process necessitates ligation of perforating vessels that supply the overlying subcutaneous tissue, rendering the flaps vulnerable to ischaemia, necrosis and infection. In an attempt to overcome this limitation, an endoscopic technique was developed to access the same myofascial layer through smaller, lateral incisions. This technique obviates the need for large subcutaneous flaps, a compromised blood supply, and the creation of a large dead space for potential seroma formation. While reducing the likelihood of wound necrosis and infection, the endoscopic technique also reduces the maximum medial advancement by 1–2 cm on each side. When release of the external oblique layer is insufficient, an additional 2–4 cm of fascial advancement can be gained by simultaneous release of the posterior rectus sheath.

Often, the posterior rectus sheath can be incised independently of external oblique release, allowing access to the retromuscular space and providing several other options for medial advancement. The traditional Rives–Stoppa–Wantz technique, developed in the 1970s, involves incising the posterior rectus sheath just lateral to the linea alba, allowing a retromuscular dissection to the lateral edge of the rectus sheath. If this does not gain sufficient medial advancement, further extension can be obtained by release of transversus abdominis (Novitsky et al 2012). The posterior rectus sheath can be incised again 1–2 cm medial to the linea semilunaris, safely preserving the laterally perforating neurovascular bundles. This release exposes the underlying transversus abdominis. Delicate transection of this muscle at its most medial insertion allows access to a plane beneath transversus abdominis and superficial to the transversalis fascia, preperitoneal fat and peritoneum. This plane of dissection can be extended laterally to psoas major, inferiorly to the space of Retzius, and superiorly beneath the diaphragm. The same preperitoneal plane can be entered directly at the linea alba and allows for lateral dissection, but does not provide the benefit of any fascial advancement/release.

In addition to the medial mobilization of fascia aided by myofascial flaps and component separations, these myofascial planes allow for mesh reinforcement with extensive overlap without exposure to intraperitoneal contents (Heller et al 2012).

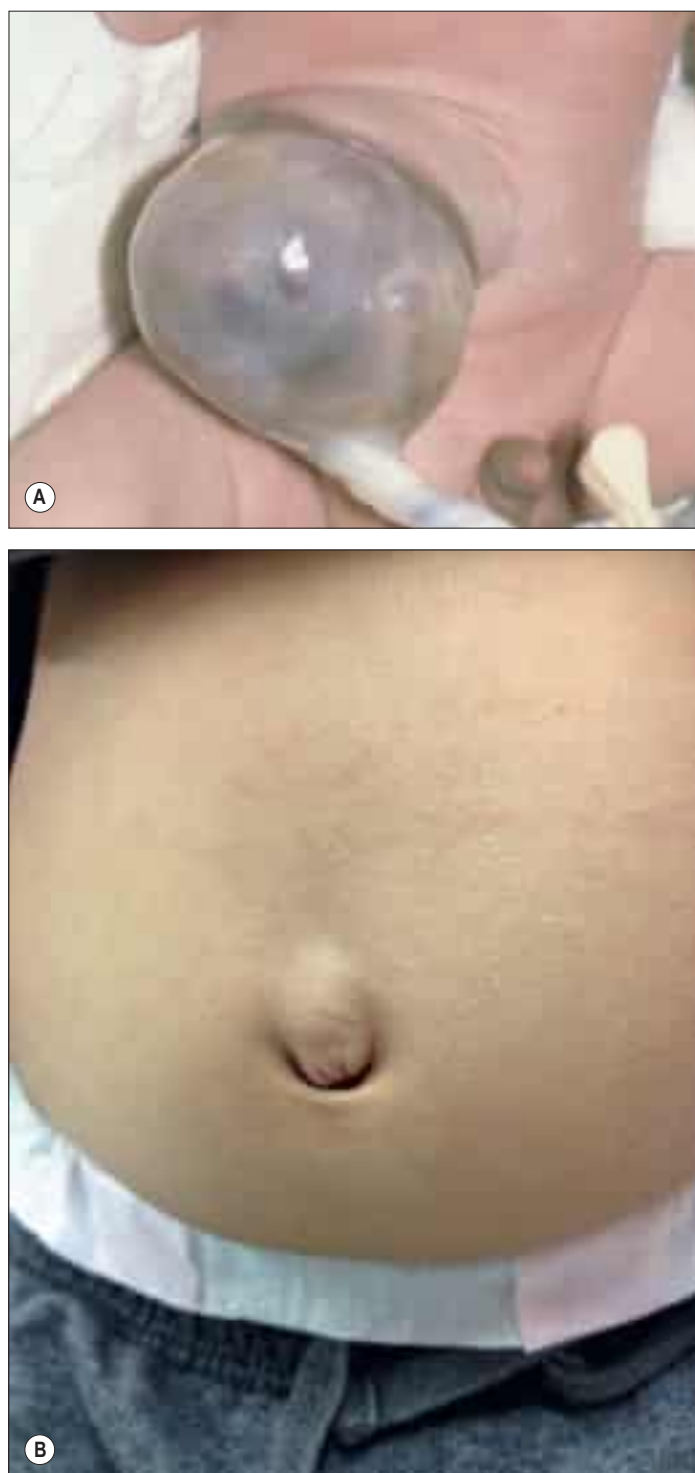


Fig. 61.17 A, A neonatal exomphalos. B, A persistent umbilical hernia in a child. (A, B, Courtesy of Professor Mark Stringer.)

ANTERIOR ABDOMINAL WALL

the region of the intersection of the linea semilunaris with the arcuate line (Skandalakis et al 2006, Larson and Farley 2002). Below this level, the aponeuroses of external and internal oblique and transversus abdominis pass anterior to rectus abdominis and the posterior rectus sheath ends. A Spigelian hernia is an interstitial hernia in that the hernia passes through a defect in the transversus and internal oblique aponeuroses but remains deep to the overlying external oblique aponeurosis. It is frequently associated with diagnostic delay.

INCISIONAL HERNIA

Incisional hernias now comprise as many as 20% of all anterior abdominal wall hernias. This complication of abdominal laparotomy incisions is due to technical failures, wound infection, and patient-related factors such as obesity, medical comorbidities and old age (Carlson et al

1995). There is currently little consensus among surgeons as to the optimum method of repairing a large incisional hernia and numerous techniques are described (Cassar and Munro 2002). Needless to say, an understanding of abdominal wall anatomy is paramount to providing a durable repair without compromising physiological function.



Bonus e-book images and table

Fig. 61.17 A, A neonatal exomphalos. **B**, A persistent umbilical hernia in a child.

Table 61.1 Distances of superior and deep inferior epigastric arteries from midline.

KEY REFERENCES

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